

Feed Additives

INTRODUCTION

Feed additives are optional diet ingredients that are not nutrients but can affect digestion, metabolism, and production. They are not required to maintain good health and high production, but production and health can be improved by some additives. Their modes of action may or may not be understood. Only additives that are approved for use in the United States (in 2019) and are the subject of peer-reviewed research involving dairy cattle are included in this review. Feed additives that also act as nutrients are discussed in the appropriate chapter. For example, buffers and chromium are discussed in Chapter 7, and niacin, biotin, and β -carotene are discussed in Chapter 8.

IONOPHORES

Ionophores are polyether antibiotics (not used in human or veterinary medicine) produced by a variety of actinomycetes that alter the flux of ions across biological membranes. Gram-negative bacteria contain a complex outer membrane and are usually unaffected by ionophores, but Gram-positive bacteria lack the outer membrane and are more sensitive to ionophores. Ionophores generally decrease the proportion of Gram-positive bacteria and increase the proportion of Gram-negative bacteria; however, changes in microbial populations are much broader than that. The ionophore monensin also causes substantial shifts in bacterial communities within those broad classes (Kim et al., 2014a,b). Furthermore, dietary ingredients and nutrients interact with monensin on altering ruminal microbial populations. In the United States, monensin and lasalocid are approved to be fed to dairy calves and growing heifers and are used as a coccidiostat for young calves (see Chapter 10) and to improve feed efficiency in growing heifers; however, in the United States, only monensin is approved for dry and lactating cows with the label claim of improving feed efficiency (solids-corrected milk/dry matter intake [DMI]). A

substantial database exists regarding metabolic (both ruminal and cow), production, and health responses to monensin, and several qualitative (McGuffey et al., 2001; Ipharraguere and Clark, 2003) and quantitative reviews (Duffield et al., 2008a,b,c; Appuhamy et al., 2013) are available. Ionophores alter production and concentrations of ruminal fermentation end products by altering ruminal bacterial populations and by altering metabolism of certain bacteria (McGuffey et al., 2001). When fed to dairy cows, methane (CH_4) production can be reduced, and the molar proportion of acetate is decreased while the molar proportion of propionate is increased. When monensin is fed to beef animals, changes in rumen volatile fatty acids (FAs) and CH_4 production are generally consistent (reviewed by McGuffey et al., 2001). However, with dairy cows fed typical diets, rumen volatile FA profiles often are not affected by monensin (Martineau et al., 2007; Oelker et al., 2009; Mathew et al., 2011). Although CH_4 production can be reduced when monensin is fed to dairy cows, the response is less and more variable than when fed to beef cattle (Appuhamy et al., 2013). Higher-forage diets and substantially greater DMI are likely reasons for the difference in ruminal responses to ionophores between beef and dairy cattle.

The increased production of propionate is one reason why cows fed monensin usually have significantly greater plasma concentrations of glucose (Duffield et al., 2008a). Production of glucose from propionate increased and tissue oxidation of glucose decreased when monensin was fed to dairy cows (Markantonatos and Varga, 2017). Monensin feeding significantly lowers concentrations of β -hydroxybutyrate and nonesterified FAs in plasma during the immediate postpartum period (Duffield et al., 2008a). In agreement with those results, cows fed monensin during the peripartum period had significantly lower risk of ketosis, mastitis, and displaced abomasum (Duffield et al., 2008c). The effects on mastitis and displaced abomasum may be indirect via a reduction in ketosis because ketosis is a risk factor for both mastitis and displaced abomasum.

Monensin often increases plasma urea concentrations in lactating dairy cows (Duffield et al., 2008a), but ionophore feeding is generally associated with decreased concentrations of ruminal ammonia (Ruiz et al., 2001). The reason for the apparently paradoxical results is unclear.

Including ionophores in diets fed to dairy cows has not markedly affected nutrient digestibility, with most responses being from 0 to about a 3 percent increase in dry matter (DM), organic matter, or energy digestibility (Haimoud et al., 1995; Knowlton et al., 1996; Plaizier et al., 2000; Morris et al., 2018b; Tebbe et al., 2018). The mode of action is not clear but may be related to slightly lower DMI and altered ruminal microbial populations. Monensin also has very modest effects on feeding behavior (more frequent and more meals per day) in peripartum cows (Mullins et al., 2012), which may help stabilize the rumen. Effects of ionophore on digestibility of carbohydrates (e.g., neutral detergent fiber [NDF] and starch) in dairy-type diets are small and inconsistent; however, apparent protein digestibility is often increased with ionophores (Plaizier et al., 2000; Ruiz et al., 2001; Benchaar et al., 2006; Martineau et al., 2007; Morris et al., 2018b). Ionophore feeding has increased flow of feed protein out of the rumen (Haimoud et al., 1995), and if that protein is more digestible than microbial protein, that could increase protein digestibility. Ionophores also increase absorption of certain minerals (see Chapter 7).

A meta-analysis using data from 36 papers and more than 9,600 cows concluded that monensin reduced DMI by 2.3 percent, increased milk yield by 2.3 percent, and reduced both milk fat and milk protein percentage but increased protein yield (did not affect fat yield). Energetic efficiency (milk energy plus energy change in body divided by energy intake) was significantly increased by 2.5 percent (Duffield et al., 2008b). The DMI equations do not include a monensin term because the database used for those equations was not adequate to separate a monensin effect. However, if monensin is included in the diet within the range of approximately 200 to 450 mg/d (approximate range in Duffield et al., 2008b), the model increases the digestible energy concentration of the diet by 2 percent (Fairfield et al., 2007) and reduces CH₄ energy by 5 percent (Odongo et al., 2007), which results in an approximate increase in metabolizable energy of 2.7 percent, which equals a 2.5 percent increase in net energy for lactation (NEL). Improvement in efficiency is likely caused by a combination of factors described above, including slightly improved digestibility, increased propionate production, and reduced CH₄. Based on a meta-analysis, monensin often reduces milk fat percentage; however, significant heterogeneity among studies was found, and numerous individual studies report no effect of monensin on milk fat percentage. Generally, monensin is more likely to reduce milk fat percentage when fed in diets with greater concentrations of C18:3 (Duffield et al., 2008b). Increasing dietary concentrations of both C18:1 and C18:2 linearly reduced milk fat percentage, but no interactions

between monensin feeding and type or amount of dietary unsaturated FA were observed (He et al., 2012). In addition, Mathew et al. (2011) reported that feeding monensin reduced milk fat concentration, but the effect occurred whether supplemental fat (mix of 18-carbon unsaturated FAs) was fed or not. Including almost 30 percent of diet DM as distillers grains caused significant milk fat depression, and that was exacerbated by addition of monensin (Morris et al., 2018b). Diets were high in C18:2 but also high in sulfur and had a low dietary cation anion difference, which can also reduce milk fat (see Chapter 7). The causes of the variability in milk fat response to monensin have yet to be explained fully. Monensin supplementation often increases the concentrations of trans FAs including trans-10, cis-12 conjugated linoleic in milk with or without an effect on milk fat concentrations (da Silva et al., 2007; Oelker et al., 2009; Mathew et al., 2011; He et al., 2012; Morris et al., 2018a).

Growing Heifers

When fed to beef animals consuming high-starch diets, ionophores usually decrease feed intake but have little effect on daily gain, thereby improving feed conversion efficiency (reviewed by NASEM, 2016). In studies specific to dairy heifers, feed intake is not significantly reduced by ionophore supplementation (Baile et al., 1982; Meinert et al., 1992; Steen et al., 1992), and average daily gain or efficiency of feed utilization is often increased, but the differences have not always been significant (Baile et al., 1982; Meinert et al., 1992). Similar results are observed when beef animals are fed ionophores in high-forage diets (Bretschneider et al., 2008). Type and dose of ionophore, forage quality, and interactions between those factors can affect growth responses to ionophores. A reduction in days to conception (Baile et al., 1982) or age at first breeding and age at first calving (Meinert et al., 1992) has been reported. When ionophores were fed to recently weaned calves (approximately 12 weeks old), no or negative effects on growth rate have been reported (Cabral et al., 2013; Chapman et al., 2016).

YEAST AND DIRECT-FED MICROBIALS

A direct-fed microbial (DFM) as defined by the U.S. Food and Drug Administration is a feed additive that contains viable microorganisms. Based on descriptions by the Association of American Feed Control Officials (AAFCO), yeast, if viable, is considered a DFM. Yeast culture products and certain other yeast products (e.g., brewer's yeast) do not contain appreciable, if any, viable cells and are not DFMs. *Saccharomyces cerevisiae* is the most commonly fed yeast DFM; bacterial DFMs include various *Propionibacterium* and *Lactobacillus* species or strains including *Enterococcus faecium*, *Prevotella bryantii*, and *Megasphaera elsdenii*. Observed responses to DFMs and yeast products include greater milk yields, altered milk composition, greater feed

intake, greater feed efficiency, altered ruminal organic acid profiles, and higher ruminal pH. Proposed modes of action of yeast products and DFMs include altering ruminal bacterial population, including increased number of lactic acid-using bacteria, synthesis of growth factors or vitamins, reduction of oxygen concentrations within the rumen, and increased overall microbial activity and mass of microbes (Yoon and Stem, 1995; Seo et al., 2010). In nonruminants, DFMs have effects within the intestines, and this may also occur in ruminants.

Two meta-analyses have been conducted to quantify production responses by dairy cows fed yeast products (Desnoyers et al., 2009; Poppy et al., 2012). Both meta-analyses included studies that fed *S. cerevisiae* culture products. One analysis included only studies evaluating production responses by dairy cows fed a yeast product from a single company (Poppy et al., 2012). The other analysis included yeast from multiple companies fed to ruminants (Desnoyers et al., 2009). Several of the same studies were included in both analyses. Both analyses concluded that yeast culture increases milk yield and yield or concentration of milk fat. Yeast increased milk protein yield in one meta-analysis (Poppy et al., 2012) but not in the other. In one meta-analysis (Desnoyers et al., 2009), yeast culture increased DMI, but in the other analysis (Poppy et al., 2012), yeast increased DMI in early lactation (<70 days in milk) but reduced it in later lactation. Milk yield and, to a lesser extent, DMI responses were positively related to dose of yeast (Desnoyers et al., 2009). Cows fed yeast had slightly but significantly higher rumen pH and volatile FA concentrations and lower lactic acid concentrations. Total tract organic matter was increased significantly by about 1.1 percent.

Several species or classes of bacteria have been evaluated as potential DFM for cattle, but the number of studies for each type of bacteria is limited and not adequate for a meta-analysis. In addition, bacterial DFMs are often fed in combination with yeast products, which makes attributing responses to a specific DFM impossible. An extensive qualitative review on responses to bacterial DFMs is available (McAllister et al., 2011). Although bacterial DFMs may have multiple modes of action, they are often classified as lactic acid producers or lactic acid utilizers (Seo et al., 2010). Lactic acid-producing bacteria that have been fed to dairy cows include *Enterococcus faecium* and various *Lactobacillus* spp. One proposed mode of action is that these bacteria will produce low amounts of lactic acid constantly over time, which will stimulate growth of lactic acid-using bacteria. When a large influx of fermentable carbohydrate into the rumen occurs, the greater population of lactic acid-using bacteria will help attenuate rumen lactic acid concentrations and ruminal pH. Some data are available showing that at least regarding ruminal pH, this may occur (Nocek et al., 2002). When *E. faecium* was fed in combination with yeast products, feed intake and milk yield and milk component yields were increased (Nocek et al., 2003); however, this effect may be from the yeast, the bacteria, or their combination. Other

suggested modes of action for lactic acid-producing bacteria include antibacterial activity against specific bacteria and alteration of intestinal microbiome resulting in improved immune response (McAllister et al., 2011). Lactic acid utilizers that have been evaluated as DFM include *Megasphaera elsdenii* and various *Propionibacterium* spp. Wild-type *M. elsdenii* is a major lactic acid utilizer in the rumen, and various strains have been fed. Dairy cows dosed with *M. elsdenii* have or tended to have higher concentrations of ruminal propionate and lower acetate to propionate ratios, but milk production, milk composition, feed intake, and ruminal pH have not been affected (Hagg et al., 2010; Aikman et al.,

2011). *Propionibacteria* can ferment lactate into propionate, but they can also produce propionate from alternative pathways. Increased ruminal propionate production could increase glucose synthesis, which could increase milk yield or metabolic efficiency. When *propionibacteria* were fed to dairy cows, efficiency or milk yield was increased in three of four studies (Francisco et al., 2002; Stein et al., 2006; Raeth-Knight et al., 2007; Weiss et al., 2008). In nonruminants and calves, DFM can modify microbial populations within the intestine, which can reduce certain diseases and enhance efficiency. This area has not been researched extensively in ruminants, but such research could lead to a better understanding of how DFM works.

SILAGE INOCULANTS

The effects of silage inoculants on the fermentation of silage are beyond the scope of this book (see Muck et al., 2018, for a discussion on this topic); however, since the inoculants and its end products are ultimately consumed by cows, they can be considered a feed additive. Silage inoculants can be broadly classified as homofermentative lactic acid bacteria (LAB) and obligate heterofermentative LAB. Based on a meta-analysis (Oliveira et al., 2017), cows fed silage inoculated with homofermentative LAB (this classification also includes facultative heterofermentative LAB, but they produce essentially only lactic acid) produced more milk than cows fed uninoculated silage likely because of greater DMI. Digestibility and feed efficiency were generally unaffected. The mode of action is unclear, but lower concentrations of some potentially hypophagic compounds (e.g., butyrate or ammonia) may be involved. Based on rumen in vitro studies, silage inoculation may also alter ruminal fermentation. For example, Jalil et al. (2009) reported less CH_4 production when inoculated silage was incubated in an in vitro system compared to control silage. Muck et al. (2018) reviewed studies that evaluated cow responses to silage inoculated with obligate heterofermentative LAB (*Lactobacillus buchneri* was the only species evaluated) and concluded the DMI was not affected by inoculation with *L. buchneri*. A field study on 39 farms reported no effect on DMI or milk yields when cows were fed silage inoculated with *L. buchneri* (Kristensen et al., 2010).

ENZYMES

The exogenous enzymes used as feed additives are produced by fungi or bacteria and can have fibrolytic, proteolytic, or amylolytic activities (or any combination of those). A substantial number of studies have been conducted evaluating the value of feeding exogenous enzymes to dairy cattle, and a comprehensive list of individual papers is cited in reviews (Ortiz-Rodea et al., 2013; Adesogan et al., 2014; Meale et al., 2014; Arriola et al., 2017); newer papers not included in those reviews are also available (Daniel et al., 2016; Romero et al., 2016; Tewoldebrian et al., 2017). The most common type of enzyme additive has fibrolytic activity, and although responses are quite modest, they usually tend to increase DM and fiber digestibility (Arriola et al., 2017). Small but significant increases in feed intake and milk and milk component yields are also expected with those enzymes (Arriola et al., 2017). Although a meta-analysis indicates modest responses in digestibility and production are likely, substantial variation in responses among studies is evident. Variation in response is caused by experimental conditions (e.g., responses are less likely in Latin square-type experiments than longer-term experiments), type of animal (e.g., early lactation cows are more likely to respond than later lactation cows), enzyme type, and probably several dietary interactions (Adesogan et al., 2014). Various types of amylases have also been evaluated (DeFrain et al., 2005; Rojo et al., 2005; Tricarico et al., 2008; Klingeman et al., 2009; Gencoglu et al., 2010; Ferraretto et al., 2011; Weiss et al., 2011), and in most studies, modest improvements in digestibility, feed efficiency, or milk yield were reported. Increases in *in vivo* DM or NDF digestibility, but not starch digestibility, have been reported even though the additives did not have appreciable fibrolytic activity.

The obvious assumed mode of action is that the hydrolytic activity of the enzyme digests nutrients either while in the feed bunk or within the rumen. However, compared to the total enzymatic activity within the rumen and intestine, the amount of enzymatic activity added is minor. Beauchemin et al. (2004) and Adesogan et al. (2014) discussed potential modes of action for enzymes, and they include (1) preingestion hydrolysis, (2) continued enzymatic activity within the rumen, (3) synergistic effects with microbial enzymes, (4) enhanced bacterial attachment to feed particles, and (5) stimulation of microbial growth within the rumen. In nonruminants, some enzymes reduce viscosity of intestinal contents, thereby enhancing digestibility, but whether exogenous enzymes maintain activity postruminally is unknown.

ESSENTIAL OILS AND OTHER PHYTONUTRIENTS

Phytonutrients are plant-derived compounds that can have antimicrobial activity and direct effects on mammalian cells (Oh et al., 2017). Essential oils, a type of phytonutrient, are secondary plant metabolites that can be extracted via steam

distillation. From a nutritional standpoint, they are neither essential nor oil. The compounds often have an aroma or essence, and they are liquid and hydrophobic—hence the name essential oils. Because many of these compounds have antimicrobial activity, they have been evaluated as rumen modifiers. Extracts of plants used as seasonings in human diets such as garlic, cinnamon, oregano, rosemary, turmeric, capsaicin, cloves, and others have been studied *in vitro* and *in vivo*, and several reviews are available (Calsamiglia et al., 2007; Benchaar et al., 2008; Benchaar and Greathead, 2011; Cobellis et al., 2016). Cobellis et al. (2016) provide an extensive listing of experiments evaluating effects of numerous essential oils on *in vitro* rumen fermentation. In most studies, *in vitro* DM disappearance and production of CH₄, ammonia, and volatile FAs were reduced when essential oils were added (supplementation rates were often much higher than would be used *in vivo*). The reduction in rumen CH₄ production has potential benefits with regards to environmental impact and energetic efficiency; however, in most studies, the decrease in CH₄ production was associated with a decrease in DM disappearance that likely would mitigate any potential benefits. Some essential oils reduce ruminal populations of Archaea and protozoa, which could reduce CH₄ production. Unfortunately, those reductions frequently occur in concert with reductions in fiber-digesting bacteria (Cobellis et al., 2016).

With some exceptions, *in vitro* responses to various essential oils are reasonably consistent (e.g., reduced CH₄ production). When fed to dairy heifers (Chapman et al., 2016) or cows, responses have been inconsistent, but most studies report no effects on intake, milk production, or milk composition (Benchaar et al., 2006; Yang et al., 2007; Tassoul and Shaver, 2009; Tager and Krause, 2011; Tekippe et al., 2011, 2013; Flores et al., 2013; Hristov et al., 2013; Vendramini et al., 2016). In a few studies, milk yield (Kung et al., 2008; Ferreira de Jesus et al., 2016), milk per unit of DMI (Tassoul and Shaver, 2009; Tekippe et al., 2011; Hristov et al., 2013), and *in vivo* fiber digestibility (Benchaar et al., 2006; Tekippe et al., 2013) have been increased with essential oil supplementation. Various measures of immune function, inflammation, hepatic function, and other physiological function have not been affected to any great extent by essential oil supplementation (Drong et al., 2017a,b).

Source of essential oil, dose, and diet may affect response, but the available data are inadequate to quantify sources of variation. Duration of supplementation can affect response, but results differ among studies. Blanch et al. (2016) supplemented a mix of essential oils to cows, and it took 15 days before an increase in milk yield was observed. In another study (Klop et al., 2017), *in vivo* CH₄ production was reduced by feeding essential oils during the first 2 weeks of the experiment, but no effects were found during the next 8 weeks. Based on *in vitro* data, essential oils hold promise, but additional research is needed to identify important sources of variation affecting *in vivo* responses to essential oils.

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